Object Oriented Programming (CSEG1044)

Unit 01 – Lecture 04: Control Flow + Arrays + Strings

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Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed February 27, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed February 27, 2026).

Quick Links

Overview Key Topics Demo Summary

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Agenda

1 Overview

2 Key Topics

3 Demo / Example

4 Summary

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Learning Outcomes

- Use `if/else`, `switch`, and loops to control program flow.

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Learning Outcomes

- Use `if/else`, `switch`, and loops to control program flow.
- Create/traverse 1D and 2D arrays and solve simple array problems.

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Learning Outcomes

- Use `if/else`, `switch`, and loops to control program flow.
- Create/traverse 1D and 2D arrays and solve simple array problems.
- Explain `String` immutability and use `StringBuilder` when needed.

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Unit 01: Introduction to OOPs

- Lecture focus: Control Flow + Arrays + Strings

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Today: Roadmap

- Program control statements (selection + loops)

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Today: Roadmap

- Program control statements (selection + loops)
- Arrays (1D/2D), types of arrays, common patterns

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Today: Roadmap

- Program control statements (selection + loops)
- Arrays (1D/2D), types of arrays, common patterns
- Strings (immutability, common methods), ‘StringBuilder‘ (need-based)

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed February 27, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed February 27, 2026).

Control flow (quick)

- Selection: `if/else`, `switch`

Common pitfalls

- Off-by-one errors in loops
- Missing `break` in `switch` (unintended fall-through)

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Control flow (quick)

- Selection: `if/else`, `switch`
- Loops: `for`, `while`, `do-while`

Common pitfalls

- Off-by-one errors in loops
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Control flow (quick)

- Selection: `if/else`, `switch`
- Loops: `for`, `while`, `do-while`
- Useful keywords: `break`, `continue`

Common pitfalls

- Off-by-one errors in loops
- Missing `break` in `switch` (unintended fall-through)

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Arrays (1D/2D)

- 1D: `int[] a = new int[n];` and `a.length`

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Arrays (1D/2D)

- 1D: `int[] a = new int[n];` and `a.length`
- Traversal: `for (int i=0; i<a.length; i++)` or `for (int x: a)`

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- 2D: `int[][] m = new int[r][c];`

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- Traversal: `for (int i=0; i<a.length; i++)` or `for (int x: a)`
- 2D: `int[][] m = new int[r][c];`
- Jagged arrays: each row can have different length (use `m[i].length`)

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Strings (must-know)

- `String` is **immutable**: methods return new strings.

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Strings (must-know)

- `String` is **immutable**: methods return new strings.
- Compare content with `equals()`, not ==.

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Strings (must-know)

- `String` is **immutable**: methods return new strings.
- Compare content with `equals()`, not `==`.
- Repeated concatenation in loops is slow → use `StringBuilder`.

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Example: grades (arrays + control flow) (1/2)

```java
import java.util.Scanner;

public class GradeStats {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of students: ");
        int n = sc.nextInt();

        int[] marks = new int[n];
        for (int i = 0; i < n; i++) {
            System.out.print("Marks for student " + (i + 1)
                + ": ");
            marks[i] = sc.nextInt();
        }
```

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Example: grades (arrays + control flow) (2/2)

```
// continued...
        int max = marks[0], sum = 0;
        for (int x : marks) { sum += x; if (x > max) max =
            x; }
        double avg = sum / (double) n;
        System.out.printf("Max=%d, Avg=%.2f%n", max, avg);
        sc.close();
    }
}
```

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Mini Exercises (2–3 mins)

1. Array of length 10: what is the last valid index?

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Mini Exercises (2–3 mins)

1. Array of length 10: what is the last valid index?
2. Which compares string content: `==` or `equals`?

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Mini Exercises (2–3 mins)

1. Array of length 10: what is the last valid index?
2. Which compares string content: `==` or `equals`?
3. Which loop runs at least once: `while` or `do-while`?

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Key Takeaways

- Use selection + loops to express logic clearly (avoid off-by-one bugs).

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- Use selection + loops to express logic clearly (avoid off-by-one bugs).
- Arrays store many values of the same type; `length` guides safe iteration.

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Key Takeaways

- Use selection + loops to express logic clearly (avoid off-by-one bugs).
- Arrays store many values of the same type; `length` guides safe iteration.
- Strings are immutable; use `equals()` and `StringBuilder` when needed.

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Checkpoint Questions

1. How do you avoid off-by-one errors in array loops?

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Checkpoint Questions

1. How do you avoid off-by-one errors in array loops?
2. Why should we use `equals()` for strings?

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Checkpoint Questions

1. How do you avoid off-by-one errors in array loops?
2. Why should we use `equals()` for strings?
3. When is `StringBuilder` preferred over `String` concatenation?

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Next Unit Preview

Next: Unit 02 – Classes, Objects, and Encapsulation

- We will start writing proper classes with private fields and methods.

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Next Unit Preview

Next: Unit 02 – Classes, Objects, and Encapsulation

- We will start writing proper classes with private fields and methods.
- Bring one example of a real-world entity you want to model as a class.

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